



A Multi-product Competitive Facility Location Model for Food Purchase: Does the Store Type Matter?

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Abstract. Food supply chains face challenges due to geophysical fragmentation and distribution issues, particularly in urban retail areas. This study proposes a network of nanostores (small, family-owned retailers) to improve access to fresh food in underserved communities. The case-study is applied to the Sabana Centro region of Colombia. Using a Multi-Product Competitive Facility Location Model and discrete choice model, we capture consumer preferences and retail performance factors. The study explores how consumer retail choices impact nanostore network design and product selection. Key results show that location and price are significant factors influencing consumer choice. The model selects stores on the city outskirts to avoid saturation. The research highlights the strategic role of small retailers in ensuring food accessibility and affordability in peripheral communities.

Keywords: Competitive Facility Location · Multi-Product · Nanostores · Consumer Preferences

1 Introduction

Retail location plays a crucial role in shaping customer experiences, accessibility, and convenience (Migdadi & Abdel-Rahman, 2020). In urban planning, it influences transportation systems and sustainable development strategies (Ozuduru & Guldman, 2013). Particularly in food supply chains, retail placement and product assortment can significantly affect consumer behavior and social interactions (Mogil et al., 2022; Tuomala & Grant, 2022). Food environments typically include diverse formats such as supermarkets, wholesalers, farmers' markets, and nanostores (Boulaksil et al., 2019).

Nanostores, also known as mom-and-pop stores or corner shops, are small, family-owned outlets embedded within neighborhoods. They offer daily essentials and foster

strong community ties through trust and cultural affinity (Fransoo et al., 2017; Mejía-Argueta et al., 2019; Salinas-Navarro et al., 2024). In emerging markets, these stores are often preferred over modern retailers due to their proximity, affordable product sizes, and reliable supply (Blanchard & Lyson, 2008; Paswan et al., 2010). Properly positioned nanostores can enhance neighborhood sustainability by reducing the need for travel, promoting local economies, and supporting low-income households (Arias Jiménez, 2021; Mejía-Argueta et al., 2019). Despite their importance, nanostores face challenges such as limited economies of scale, space, and capital. Their survival is threatened by volatile environments and lack of logistical capabilities, including inventory management and supply strategies (Cannuscio et al., 2013; Kin, 2020; Rangel-Espinosa et al., 2020). Addressing these limitations is key to strengthening their role in resilient and inclusive food systems.

In this direction, this study examines competition between a proposed nanostore network offering fresh food options (fruits, vegetables, and tubers) and other retail channels in the region, such as supermarkets, existing nanostores, and fruvers (specialized fruit and vegetable retailers in Colombia). We incorporated constraints such as distance, retailer pricing, and minimum demand to a competitive facility location problem (CFLP) with product selection. We used data from the Sabana Centro region in Colombia, and the case study aims to answer the following research questions: What is the optimal location strategy for maximizing profits for a nanostore network from underserved demand? And how does product selection play a role in placing the nanostore network and serving the existing demand? From a practical perspective, this study aims to provide valuable insights into the role of nanostores in supplying and competing in the fresh food market retail and their participation in retail environments.

This paper is organized as follows: In the next section, we present our methodological approach to the study. Section 3 presents the numerical results and offers a discussion and managerial insights. Finally, Section 4 displays the study's conclusions and future work.

2 Methodology

We present a three-steps methodology for this study: First, we performed data collection and spatial analysis to generate our case study. To this, we conducted a structured survey with the households in the selected region, to gain insights into their demand patterns for the chosen items. Further, we conducted a geographical analysis to collect geolocation of the different retailers and households in the region. Then, we fit a multi-logit regression model, to assess the household's utilities associated with different retail channels. Finally, we formulate and solve a Mixed Integer Non-Linear Program (MINLP) of a Competitive Facility Location Problem (CFLP) with parameters found in the regression. The CFL approach identified the optimal locations and product selection for new retail facilities to maximize market share.

The case study was applied to the Sabana Centro region in Colombia, specifically in the municipalities of Chía and Cajicá. This region has a vast presence of nanostores among other retail formats, such as supermarkets and fruvers (da Silva-Ovando et al., 2021). In 2020, a regional survey was carried out, targeting 537 households in Chía

(288) and Cajicá (249) in collaboration with the Universidad de la Sabana and the central territorial associative scheme for administrative and special planning (RAP-E, see <https://regioncentralrape.gov.co/>). The descriptive analysis of the surveys' results was explored by (da Silva-Ovando et al., 2021). We used the data to extrapolate the potential demand using the geolocation of the surveyed households. Then, retailers, including 81 nanostores, 60 fruvers, 17 butcher's shops (i.e., specialty retailers), eight convenience/grocery stores, nine hard-discounters, and 13 chain supermarkets, were identified and mapped in the region. Fifty potential locations for the proposed nanostore network were defined according to the region's availability. Finally, the produce prices used were averaged from the price per kg of each food category per retail channel in the region. This information was collected by reviewing the prices of representatives of each category of retailers in the region.

2.1 Model Formulation

In this section, we present the model formulated to solve this case. Table 1 presents the model's objective function and restrictions, while Table 2 displays the linearization constraints applied.

Sets

- R = set of retailers indexed in r
- I = set of potential new nanostores indexed in $i, I \subseteq R$
- J = set of households indexed in j
- P = set of products indexed in p
- K = alias of I indexed in k

Parameters

- d_{ik} = Distance between potential new *nanostore* i and potential new nanostore k
- $d_{\min}$ = Minimum distance required between two potential new nanostores to be opened
- Max_{ji} = Maximum distance between household j and potential new nanostore i
- γ_{jr} = Distance between retailer r and household
- δ_j = Weekly demand of fresh food in household j for product p
- $a_{ji} = \begin{cases} 1 & \text{if } \gamma_{jr} \leq Max_{ji} \\ \frac{Max_{ji} - \gamma_{jr}}{Max_{ji} - d_{\min}} & \text{if } Max_{ji} < \gamma_{jr} \leq d_{\min} \\ 0 & \text{if } \gamma_{jr} > d_{\min} \end{cases}$
- V_{jip} = Observed utility of household j when selecting retailer i for product p , where $V_{jip} = \beta_{0(p)} + \beta_{dis\ tan\ ce\ (p)} \gamma_{ji} + \beta_{price\ (p)} P_{ip}$
- $\beta_{price\ (rp)}$ = weight of price of retailer r for product p
- $\beta_{dis\ tan\ ce\ (rp)}$ = weight of distance to choose retailer r for product p
- $\beta_{0(rp)}$ = Regression coefficient (i.e., intercept) for each retailer r for product p
- φ_{ji} = Linearization parameter of the constant, where $\varphi_{ji} = \frac{e^{V_{ij} a_{ji}}}{\sum_{r \in R/I} e^{V_{rj}}}$
- p_{rp} = Price in US\$ per kg of product p in retailer r
- c_{rp} = Unit cost in US\$ per kg of product p in retailer r

f_i = Weekly operation fixed costs in US\$ of nanostore i .

nd_{ip} = Minimum demand to cover for product p in nanostore i

Variables

$x_i = \begin{cases} 1 & \text{if the nanostore } i \text{ is open} \\ 0 & \text{otherwise} \end{cases}$

y_{jrp} : Probability of household j of choosing retailer r to purchase product p .

$\bar{y}_{jp}$: Linearized cumulative probability per household j to purchase product p .

q_{ip} = Demand captured by nanostore i per product p

$z_{ip} = \begin{cases} 1 & \text{if the product } p \text{ is sold at the nanostore } i \\ 0 & \text{otherwise} \end{cases}$

Table 1. Objective function and constraints.

$\text{Max } g = \sum_p \sum_{i \in I} (p_{ip} - c_{ip}) * q_{ip} - \sum_{i \in I} f_i x_i$		(1)
$q_{ip} = \sum_{j \in J} \delta_{jp} y_{jip}$	$\forall i \in I, p \in P$	(2)
$y_{jip} = \frac{e^{V_{jip} z_{ip} a_{ji}}}{\sum_{i \in R} e^{V_{jip} z_{ip} a_{ji}} + \sum_{r \in R/I} e^{V_{jrp} a_{jr}}}$	$\forall j \in J, i \in I, p \in P$	(3)
$d_{\min} + (Max_{ji} - d_{ik})x_i + (Max_{ji} - d_{ik})x_k \leq 2Max_{ji} - d_{ik}$	$\forall i \in I = 1, \dots, n-1$ $\forall h \in I = 1+i, \dots, n$ $\forall j \in J$	(4)
$z_{ip} \leq x_i$	$\forall i \in I, p \in P$	(5)
$\sum_{p \in P} q_{ip} \geq n d_{ip} x_i$	$\forall i \in I$	(6)
$x_i \in \{0, 1\}$	$i \in I$	(7)
$y_{jip} \in [0, 1]$	$\forall j \in J, i \in I, p \in P$	(8)
$z_{ip} \in \{0, 1\}$	$i \in I, p \in P$	(9)
$q_{ip} \in \mathbb{R}^+$	$i \in I, p \in P$	(10)

The objective function (1) maximizes the utility obtained by the potential new nanostores in the retail landscape. The first constraint (2) corresponds to the total demand of each product category per household that is captured by nanostore. Constraint (3) computes the probability of households selecting a given nanostore to get their products from using the logit model presented in Sect. 2. Constraint (4) limits the minimum distance between two potential new nanostores to be opened to avoid the cannibalization effect. Constraint (5) ensures that a product can only be sold at a nanostore if the nanostore is operational. Constraint (6) establishes that the total demand captured by nanostore across all products must reach at least a minimum threshold if the store is active. Last, constraints (7, 8, 9, 10) correspond to the domain of the variables. Constraint (3) is non-linear; therefore, we adapted the linearization algorithm proposed by (Haase & Müller, 2014). The constraint is then reformulated as displayed in constraints (11)–(14), where

we present the variable, in charge of accumulating the household probability of selecting an alternative, using the auxiliary parameter

Table 2. Linearization constraints.

$\bar{y}_{jp} + \sum_{i \in I} y_{jip} \leq 1$	$\forall j \in J, p \in p$	(11)
$y_{jip} - \frac{\varphi_{jip}}{1 + \varphi_{jip}} z_{ip} \leq 0$	$\forall j \in J, i \in I, p \in P$	(12)
$y_{jip} - \varphi_{jip} \bar{y}_{jp} \leq 0$	$\forall j \in J, i \in I, p \in P$	(13)
$\bar{y}_{jp} \in \{0, 1\}$	$\forall j \in J, p \in p$	(14)

3 Results and Discussion

In the baseline scenario, the model resulted in an objective function value of USD 26,254. A total of 17 new nanostores were opened, capturing an overall demand of 65,954 kg, (around 16% of the total demand). Four additional scenarios were developed for this analysis, with the corresponding results. Two variations were considered: first, the minimum distance required between retailers for opening was reduced from 1,000 m to 500 m (Scenario 1) and 750 m (Scenario 2), based on the assumption that households are willing to walk or drive such distances to purchase groceries. Second, demand levels were adjusted by $\pm 25\%$ (Scenarios 3 and 4) relative to the original data, accounting for possible changes due to seasonality (Hall et al., 2009), consumer preferences, and other factors (see Table 1). Despite these changes, it is assumed that households would still meet the minimum recommended intake of fruits, vegetables, and tubers as suggested by the FAO (Table 3).

Table 3. Model results – scenario variations.

Scenario	Base scenario	1	2	3	4
Minimum distance (m)	1000	500	750	1000	1000
Demand variation (%)	0%	0%	0%	−25%	25%
Objective function (USD)	26,254	42,643	30,886	17,867	34,129
Computational time (h)	0.35	1.18	1.03	0.24	0.09
New nanostores opened	17	32	21	15	18
Total demand captured (Kg)	65,954	108,210	77,844	45,689	84,776
% of total demand captured	16%	27%	19%	15%	17%

Comparative analysis across scenarios reveals notable differences in demand capture, cost-efficiency, and operational scale. Figure 1 summarizes the most frequent locations

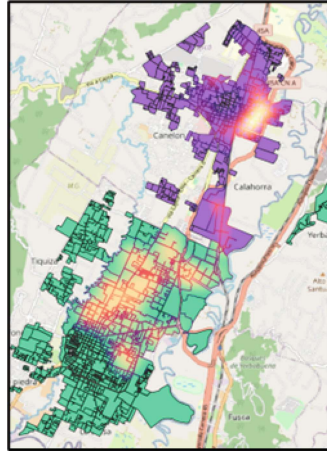


Fig. 1. Heatmap of most frequently opened nanostores across scenarios in Chía (green) and Cajicá (purple).

for nanostore openings across each scenario. It illustrates a tendency to establish nanostores primarily on the outskirts of both municipalities, avoiding downtown areas and particularly located in areas where other retail channels are absent. Scenario 1, which allowed for the shortest minimum distance between retailers (500 m), achieved the highest total demand captured (108,210 kg), covering 27% of the total estimated demand. This was possible by opening 32 nanostores and required the highest computational time (1.18 h) and objective cost (USD 42,643). Scenario 3, with a 25% demand reduction, achieved the lowest demand capture (45,689 kg or 15%) and cost (USD 17,867), making it the most conservative configuration in terms of both demand and investment. Interestingly, Scenario 4, despite its increased demand assumption (+25%), captured only 17% of the demand with a higher cost (USD 34,129), indicating diminishing returns beyond a certain demand threshold. From a product-level view, all scenarios show relatively balanced distribution among fruits, vegetables, and tubers, though Scenario 1 outperforms consistently across all categories. These findings suggest that decreasing spatial restrictions can significantly enhance coverage, but cost-benefit considerations must weigh heavily in decision-making.

4 Conclusions and Future Work

This study presented a multi-product competitive facility location model applied to fresh food retail distribution in underserved urban areas, with a specific focus on the Sabana Centro region in Colombia. By integrating discrete choice modeling into the location-allocation framework, the model effectively captured consumer preferences for store type, price, and distance. The results confirmed that nanostores can play a significant role in expanding food access, especially when spatial constraints are relaxed. The scenario with the lowest minimum distance between retailers (500 m) showed the highest demand capture (27%) and the most balanced product distribution, although it required a greater

number of openings and operational costs. The findings highlight a trade-off between coverage and efficiency. While wider deployment of nanostores increases accessibility and demand fulfillment, it also elevates costs and complexity. Conversely, conservative configurations reduce operational burden but risk under-serving demand. Moreover, product-level analysis revealed consistent consumer interest across fruits, vegetables, and tubers, suggesting a need for diverse assortments regardless of the spatial or demand scenario.

Future work should address several dimensions. First, the model can be extended to include logistics and inventory decisions, incorporating dynamic demand and restocking constraints. Second, further empirical validation could be conducted using real-time transaction data or longitudinal studies to track consumer adaptation over time. Third, future research may explore partnerships between nanostores and larger retail actors to improve supply chain resilience and economies of scale. Finally, expanding the model to other regions and retail typologies could test its generalizability and support public policy aimed at enhancing urban food security through inclusive retail networks.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], as well as a URL [5].

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